

WELL COMPLETION REPORT (SYNTHETIC)

OIL-LKW-008 - Lakwa-008

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000088
Field / District	Lakwa / Charaideo
Block	PEL-ASSAM-04
Basin	Assam-Arakan
Formation	Kopili (KOP)
Spud / Completion	2020-04-16 / 2020-10-11
Measured / TVD	3398 m / 3358 m
Trajectory	Vertical; inc 3.2 deg; az 42.5 deg
Formation Interval	1718-2909 m
Lithology	Calcareous shale and siltstone
Pore Pressure Reference	3977 psi
Fracture Gradient	15.97 ppg
Mud System	KCl Polymer WBM
NWIS Well Risk	Moderate (44.4/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	849	Surface
9 5/8 in casing	1970	Intermediate
7 in liner/casing	2990	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2020-05-15 704 m	Pressure Spike	Critical	Standpipe pressure increased rapidly at nearly stable flow. Mitigation: Reduce pump rate, verify restriction and restore circulation gradually. Lesson: Correlate pressure anomalies with flow before escalating risk.
2020-08-11 2326 m	Mud Loss	Critical	Intermittent circulation losses developed while drilling ahead. Mitigation: Spot graded LCM pill and circulate at reduced rate. Lesson: Use conservative hydraulics when re-entering equivalent loss-prone interval.
2020-06-23 1452 m	Mud Loss	High	Intermittent circulation losses developed while drilling ahead. Mitigation: Optimize hydraulics and maintain conservative ROP through loss interval. Lesson: Track flow balance and pit volume trend before losses become severe.
2020-09-05 2772 m	Pressure Spike	High	Standpipe pressure increased rapidly at nearly stable flow. Mitigation: Use pressure trend thresholds referenced to depth and flow rate. Lesson: Correlate pressure anomalies with flow before escalating risk.
2020-05-20 811 m	Mud Loss	Medium	Partial returns observed at shale shaker. Mitigation: Reduce ECD, treat losses with LCM and confirm stable returns. Lesson: Track flow balance and pit volume trend before losses become severe.
2020-05-21 835 m	Cementing Issue	Medium	Cement returns were below planned percentage. Mitigation: Verify cement placement using pressure and volume balance. Lesson: Plan conservative displacement rate in weak formations.
2020-06-14 1284 m	Stuck Pipe	Medium	Hookload became erratic during pipe movement. Mitigation: Reduce ROP and increase circulation before high-angle interval. Lesson: Rising torque plus falling ROP should trigger hole-cleaning action.

Completion Summary

The synthetic well reached 3398 m MD in the Kopili interval. The NWIS knowledge record links 12 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.